#Aviation Meteorology

Chapter Three: Turbulence

Objectives

- To Define Turbulence
- To Identify Different Types Of Turbulence
    - ✓ Mechanical Turbulence
    - ✓ Convective Turbulence
    - ✓ Frontal Turbulence
- To know the Effects of boundary-layer Turbulence on Take-offs and landings of aircraft

Turbulence


What is turbulence?
A sudden, violent shift in airflow
Causes:
•Wind
•Storms
•Jet stream
•Objects near the plane
(particularly mountain ranges)
fast wind
Aircraft can drop or change altitude suddenly
slow wind
news

Mechanical Turbulence

- Friction between the air and the ground, especially *irregular terrain* and man-made obstacles, causes eddies and turbulence in the lower levels.
- The intensity of this eddy motion depends on *the strength of the surface wind, the nature of the surface* and *the stability of the air.*

- In strong winds, even hangars and large buildings cause eddies that can be carried some distance downwind.
- Strong winds are usually quite gusty; that is, they fluctuate rapidly in speed.
- Sudden increases in speed that last several minutes are known as squalls and they are responsible for quite *severe turbulence*.

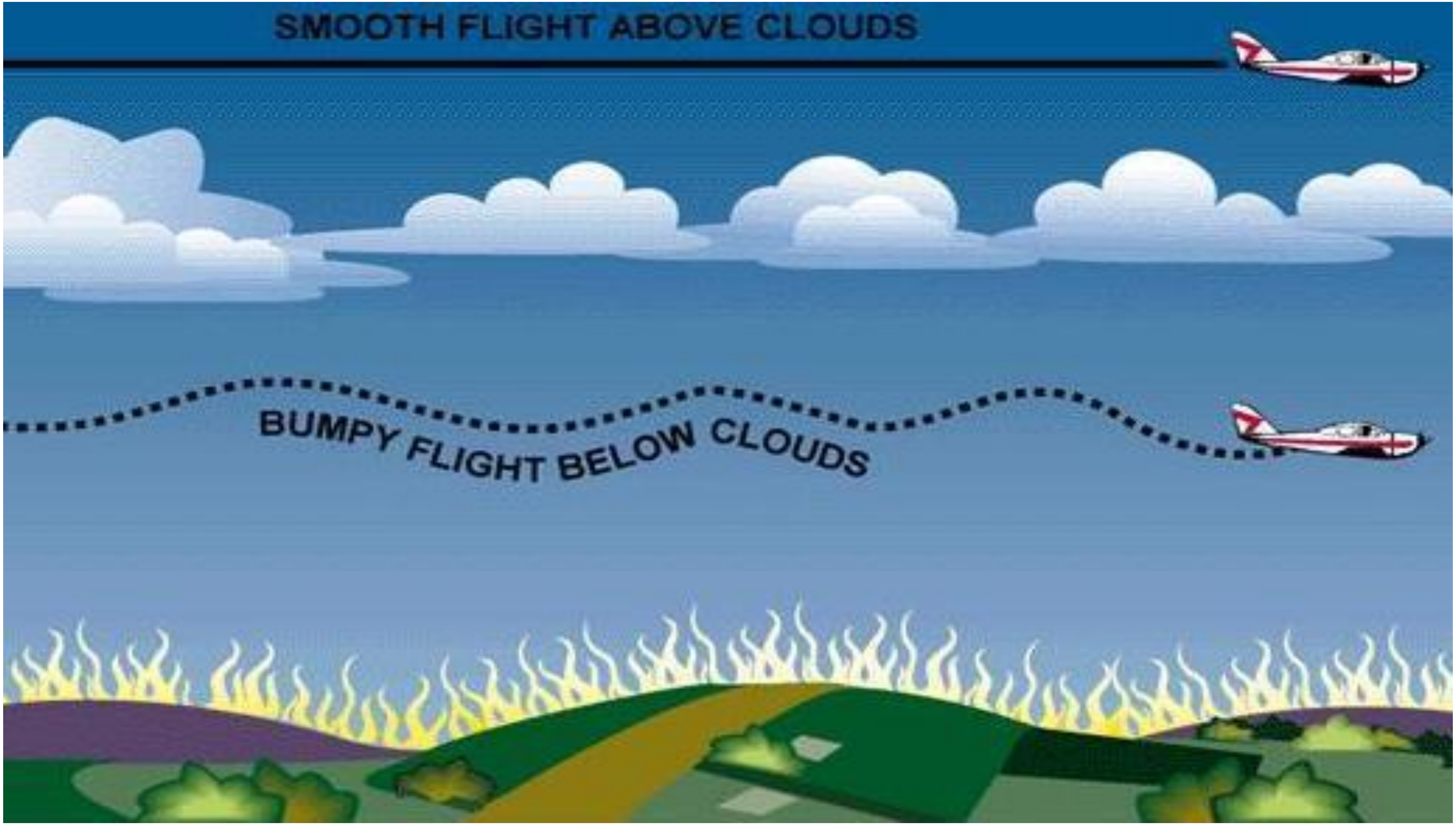
Thermal (Convective) Turbulence

- Turbulence can also be expected on warm summer days when the sun heats the earth's surface unevenly.
- Certain surfaces, such as barren ground, rocky and sandy areas, are heated more rapidly than are grass covered fields and much more rapidly than is water.
- Isolated convective currents are therefore set in motion with warm air rising and cooler air descending, which are responsible for bumpy conditions as an airplane flies in and out of them.

- Turbulence extends from the base to the top of the convection layer, with smooth conditions found above. if cumulus, towering cumulus or cumulonimbus clouds are present, the turbulent layer extends from the surface to cloud tops.
- *Turbulence intensity increases as convective updraft intensity increases*. In weather conditions when thermal activity can be expected, many pilots prefer to fly in the early morning or in the evening when the thermal activity is not as severe.

- *Favorable conditions* for dry convection include warm surface temperatures, uneven surface hearing, and steep surface-based lapse rates.
- Convective currents are often strong enough to produce air-mass thunderstorms with which severe turbulence is associated.
- Turbulence can also be expected in the lower levels of a cold air mass that is moving over a warm surface. Heating from below creates unstable conditions, gusty winds and bumpy flying conditions.

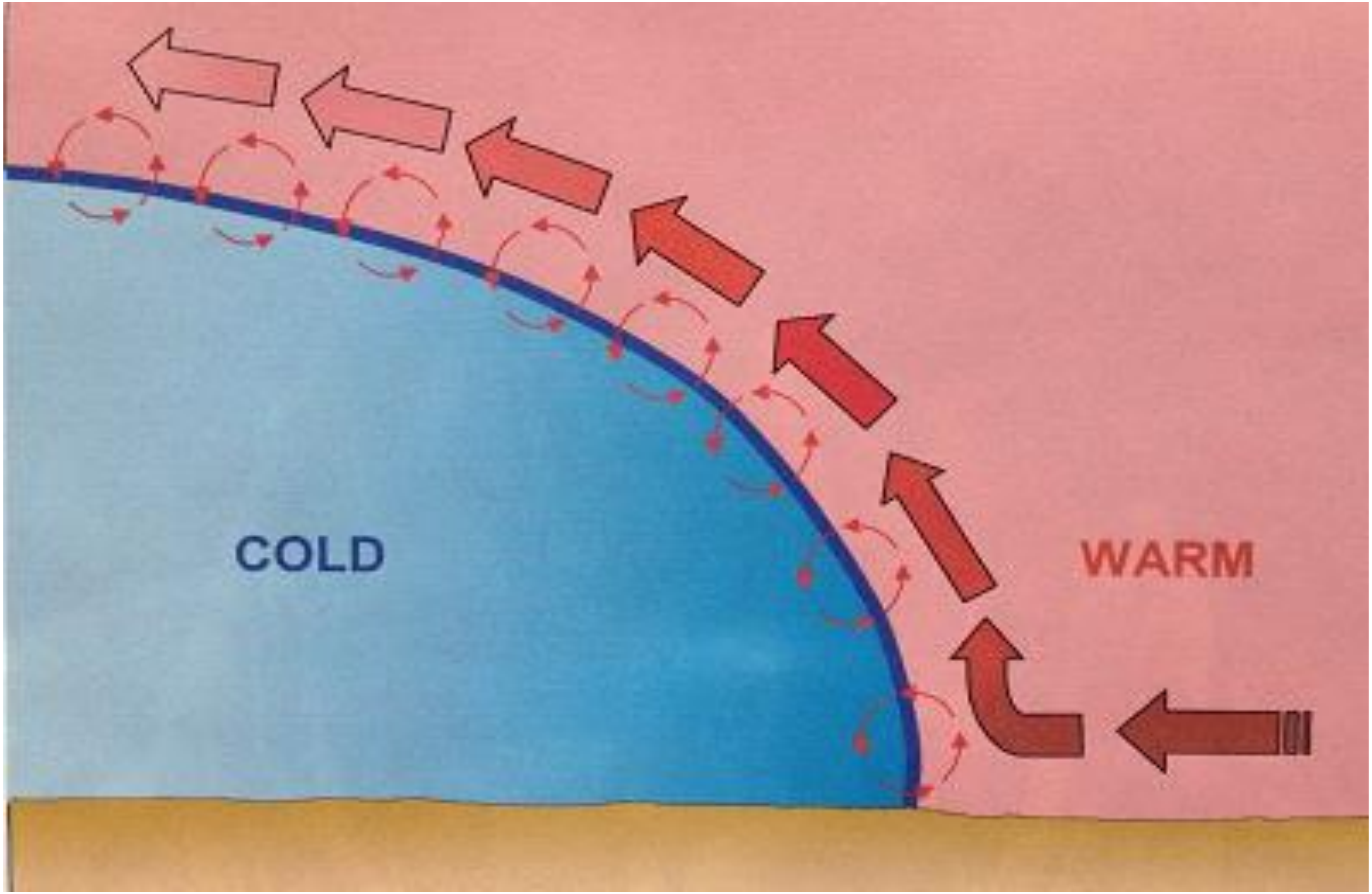
- Thermal turbulence will have a pronounced effect on the flight path of an airplane approaching a landing area.
- The airplane is subject to convective currents of varying intensity set in motion over the ground along the approach path.
- These thermals may displace the airplane from its normal glide path with the result that it will either overshoot or undershoot the runway.


SMOOTH FLIGHT ABOVE CLOUDS
BUMPY FLIGHT BELOW CLOUDS

- In an atmosphere where *hydrostatic instability* is present, convective turbulence is more likely to *occur*, especially when there is a heat source at the surface (e.g., warm ground or water).
- The heating at the surface causes air to rise, and if the atmosphere is hydrostatically unstable, this rising motion can become turbulent.
- In summary, ***hydrostatic instability*** can enhance convective turbulence by allowing for vertical motion in the atmosphere, which, when triggered by local heating or other mechanisms, leads to the development of turbulent eddies and convective currents.

Frontal Turbulence

- The *lifting of the warm* air by the sloping frontal surface and *friction between the two opposing air masses* produce turbulence in the frontal zone.
- This turbulence is most marked when the warm air is moist and unstable and will be extremely severe if thunderstorms develop.
- Turbulence is more commonly associated with cold fronts but can be present, to a lesser degree, in a warm front as well.


COLD
WARM

Wind Shear

- Wind shear is the change in wind direction and/or wind speed over a specific horizontal or vertical distance.
- Atmospheric conditions where wind shear exists include: areas of temperature inversions, along troughs and lows, and around jet stream.
- When the change in wind speed and direction is *pronounced*, quite severe turbulence can be expected.

- Temperature inversions are zones with vertical wind shear potential.
- The greatest shear, and thus the greatest turbulence, is found at the tops of the inversion layer.
- Turbulence associated with temperature inversions often occur due to radiational cooling, which is nighttime cooling of the Earth's surface, creating a surface-based inversion.

- Turbulence associated with lows and troughs is due mainly to horizontal directional and speed shear.
- Turbulence is generally found along troughs at any altitude, within lows at any altitude, and poleward of lows in the mid and upper altitudes.

CLEAR AIR TURBULENCE(CAT)

- Clear-air turbulence (CAT) occurs in clear skies when there is a significant change in wind speed or direction with altitude (vertical wind shear), and this is stronger than the stability or stratification of the atmosphere.
- The interaction between these factors can lead to the development of turbulent eddies or shearing layers, causing turbulence even when there are no visible signs such as clouds.

- Clear-air turbulence (CAT) is generated in clear skies when the vertical wind shear is stronger than the stratification.
- It is called "clear air turbulence" because it typically occurs in clear air, without any visual cues such as clouds.
- Unlike turbulence associated with thunderstorms, clear air turbulence doesn't have easily identifiable indicators.
- CAT is commonly encountered at cruising altitudes, usually above 15,000- 20,000 feet (6,000 meters).

Effects of boundary-layer Turbulence on Take-offs and landings of aircraft

- Boundary-layer turbulence can have various effects on aircraft during take-offs and landings.
- The boundary layer refers to the layer of air adjacent to the Earth's surface, and turbulence in this layer can be caused by factors such as *wind shear*, *temperature gradients*, and *surface roughness*.
- Here are some effects of boundary-layer turbulence on aircraft operations

Take-offs:

- ***Lift and Control:*** Turbulence in the boundary layer can affect the lift generated by the aircraft's wings. Sudden changes in airspeed and lift due to turbulence can make the take-off phase challenging.

- ***Rough Ride:*** Turbulence during take-off can result in a rough ride for passengers and crew. This is particularly true during the initial climb phase when the aircraft is still in close proximity to the ground.

- ***Windshear:*** Boundary-layer turbulence is often associated with windshear, which is a rapid change in wind speed and direction. Windshear during take-off can impact the aircraft's performance, especially during the critical initial climb phase.

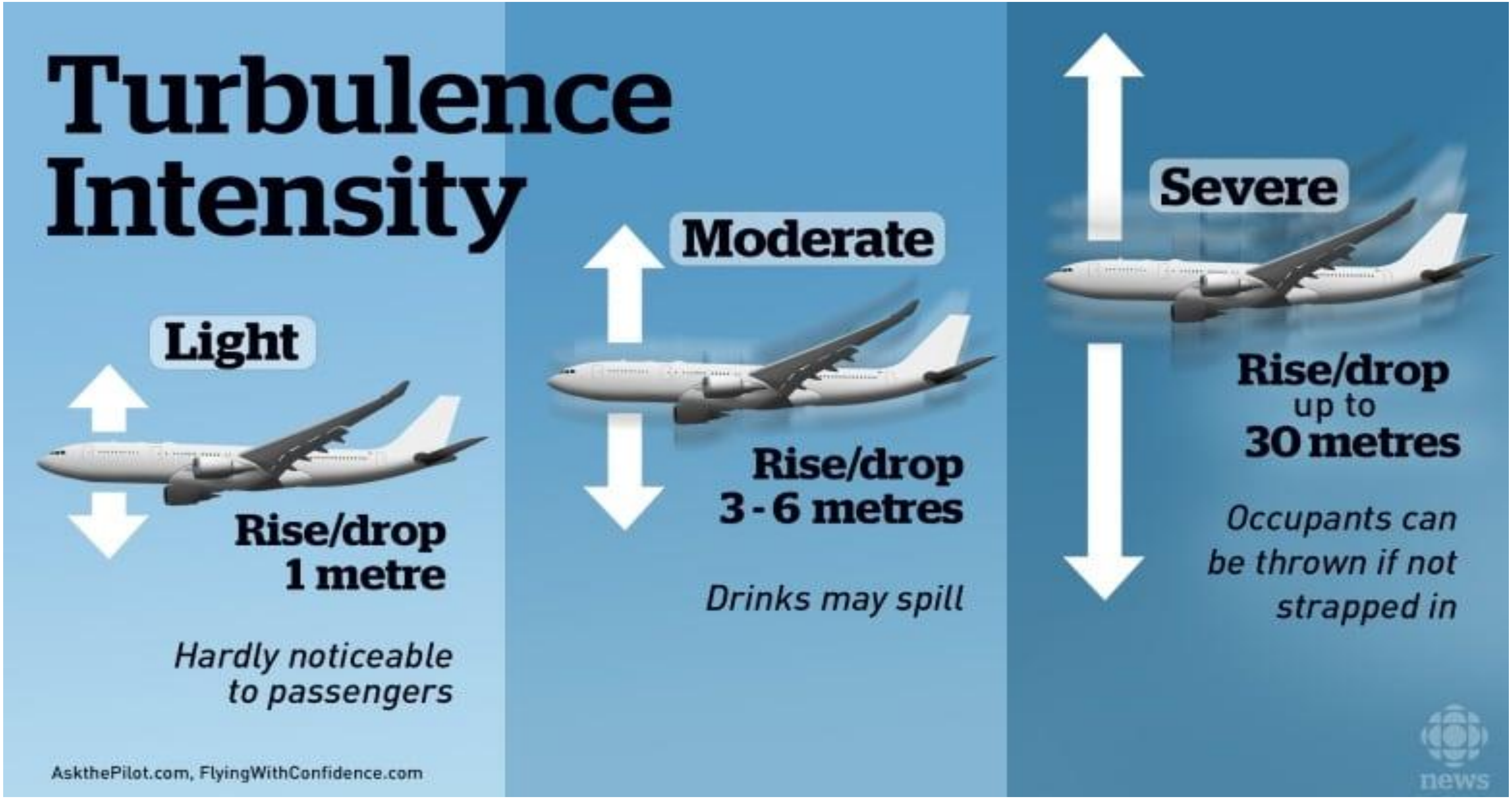
Landings

- ***Approach and Flare:*** Turbulence in the boundary layer can affect the approach and landing phase. Pilots may experience changes in the aircraft's altitude, airspeed, and descent rate, requiring careful adjustments during the approach and flare for a safe landing.

- ***Crosswinds:*** Boundary-layer turbulence can contribute to crosswind conditions during landing. Crosswinds can make the landing challenging, requiring skilled pilot techniques to align the aircraft with the runway and touch down safely.

- ***Gusts and Bumps:*** Turbulence close to the ground can result in gusty and bumpy conditions during the final approach and landing rollout. Pilots need to be prepared for these conditions and make appropriate control inputs to ensure a smooth landing.

Turbulence Intensity


Turbulence Intensity
Light
Rise/drop 1 metre
Hardly noticeable to passengers
Moderate
Rise/drop 3 - 6 metres
Drinks may spill
Severe
Rise/drop up to 30 metres
Occupants can be thrown if not strapped in
AskthePilot.com, FlyingWithConfidence.com
news

Turbulence intensity

- In reporting turbulence, it is usually classed as light, moderate, severe or extreme.

- **Light turbulence** momentarily causes slight changes in altitude and/or attitude or a slight bumpiness.
  - Occupants of the airplane may feel a slight strain against their seat belts.

Turbulence intensity

- ❖**Moderate** turbulence is similar to light turbulence but somewhat more intense.
  - ✓There is, however, no loss of control of the airplane.
  - ✓Occupants will feel a definite strain against their seat belts and unsecured objects will be dislodged.

Turbulence intensity

- ❖ **Severe turbulence** causes large and abrupt changes in altitude and/or attitude and, usually, large variations in indicated airspeed.
  - ✓ The airplane may momentarily be out of control.
  - ✓ Occupants of the airplane will be forced violently against their seat belts.
- ❖ In **extreme turbulence**, the airplane is tossed violently about and is impossible to control. It may cause structural damage.

Turbulence intensity

- **Chop** is a type of turbulence that causes rapid and somewhat rhythmic bumpiness.

Turbulence Intensity Classification	
Intensity	**Effect**
Light	Slight erratic changes in altitude and/or attitude
Moderate	Change in altitude and/or attitude, but the aircraft remains in positive control at all times
Severe	Large, abrupt changes in altitude and/or attitude. Aircraft may be momentarily out of control
Extreme	Aircraft is violently tossed about and practically impossible to control. May cause structural damage.

Turbulence and Thunderstorms

- Turbulence, associated with thunderstorms, can be extremely hazardous, having the potential to cause overstressing of the aircraft or loss of control.
- Thunderstorm vertical currents may be strong enough to displace an aircraft up or down vertically as much as 2000 to 6000 feet.

- The greatest turbulence occurs in the vicinity of adjacent rising and descending drafts. Gust loads can be severe enough to stall an aircraft flying at rough air (maneuvering) speed or to cripple it at design cruising speed.
- Maximum turbulence usually occurs near the mid-level of the storm, between 12,000 and 20,000 feet and is most severe in clouds of the greatest vertical development.

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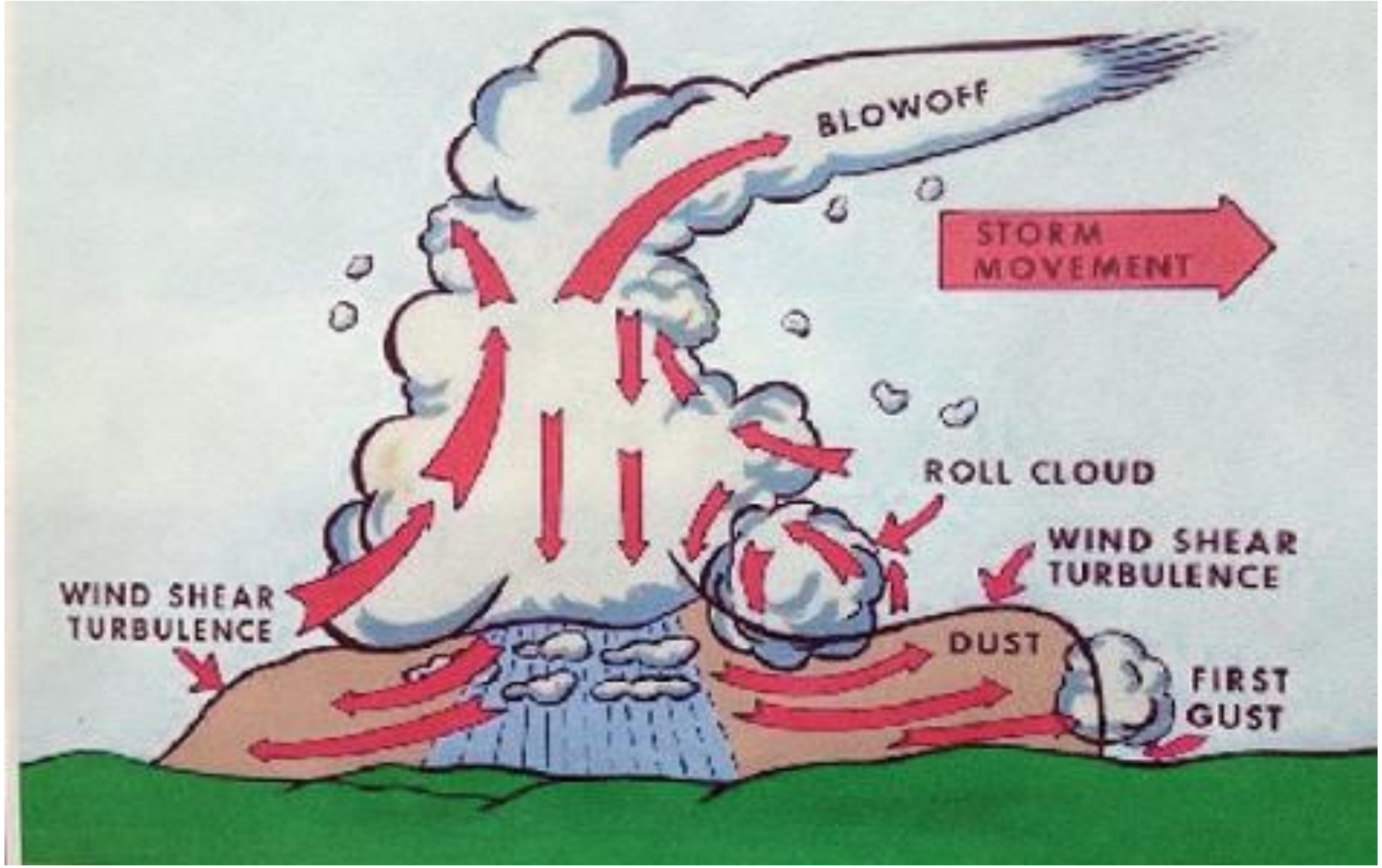
Turbulence and Thunderstorms



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Thanks !